How the three ensembles come to agree: an exact study of a two-dimensional quantum lattice gas

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Exact diagonalization of the hard-core Bose–Hubbard model

Abstract. Statistical mechanics asserts that the microcanonical, canonical, and grand-canonical ensembles agree in the thermodynamic limit. For a system of a handful of particles the agreement is only approximate, and how good the approximation is becomes a question one can answer by direct computation. We diagonalize the two-dimensional hard-core Bose–Hubbard model completely—every eigenvalue in every particle-number sector—on square lattices of 4, 9, and 16 sites, and build all three ensembles from the one spectrum. Reading the temperature and chemical potential off the microcanonical entropy by finite differences turns out to be unreliable on spectra this small and this sharply bounded, so we track equivalence instead through quantities that are smooth sums over the spectrum: the relative energy fluctuation $\sigma_E/|\langle E\rangle|$ and the relative particle-number fluctuation $\sigma_N/\langle N\rangle$, both of which fall as $1/\sqrt{M}$ when the ensembles converge. For the free gas and for repulsion they do fall, with fits close to that rate. Attraction is different. The number fluctuation stays large, and for $V/t = -3$ its finite-size exponent on these three lattices is positive—the canonical-to-grand mismatch does not shrink with the lattice, which is what a lattice gas on the verge of phase separation should do. Sweeping V/t from -3 to $+3$ turns this into a single monotonic curve in the number channel. Repeating the scaling at every temperature from $T/t = 0.2$ to 10 confines the anomaly to low temperature: at $V/t = -3$ the exponent is positive only below $T \approx 1.9\,t$, close to the $T_c \approx 1.70\,t$ at which the hopping-free gas, an Ising model, phase separates. Mapping the model onto an XXZ ferromagnet explains why $V/t = -2$, its isotropic Heisenberg point, never turns positive, and predicts its low-temperature number fluctuation exactly. Finally, we redo the scaling on one geometry: tilted bipartite clusters of 8 to 20 sites, reached with translation, point-group, and particle–hole symmetry. There every coupling gives exactly -0.500 at high temperature and the free gas -0.47 at $T = t$, while at $V/t = -3$ the exponent is 0.00: the number fluctuation does not grow, it stops shrinking, as the Ising picture predicts.

1 Introduction

Write down a box of gas three different ways—fixed energy, fixed temperature, fixed chemical potential—and thermodynamics says you get the same answer. The statement is exact only when the number of particles runs to infinity. Everything actually computed, measured in a trap, or simulated on a lattice has a finite number of particles, and then the three descriptions differ. The size of that difference, and the way it shrinks as the system grows, is the subject of this paper.

We work with the hard-core Bose–Hubbard model on a square lattice: bosons that hop between neighboring sites with amplitude t, repel or attract neighbors with energy V, and are forbidden from sharing a site. It is a quantum lattice gas in the plain sense of the phrase—a gas of quantum particles sitting on a lattice—and it is realized by cold bosonic atoms in an optical lattice in the strongly interacting limit. (It has nothing to do with the "quantum lattice-gas automaton," a cellular-automaton algorithm that shares the name.) Two features make it a good test bed. The local Hilbert space is two states per site, so the whole spectrum is reachable by exact diagonalization for small lattices, and the interaction can be dialed from repulsive through free to attractive without changing anything else.

The plan is ordinary but the order matters: build the Hamiltonian sector by sector, check it against anything that can be computed another way, diagonalize, and only then assemble ensembles and compare. One result of that discipline is negative and worth stating early. The textbook route to the microcanonical temperature—differentiate $S = \ln\Omega$ with respect to energy—does not survive contact with a spectrum of a few thousand discrete, bounded levels. We show why below and route around it.

2 Model

On an $L \times L$ lattice with $M = L^2$ sites,

$$\hat{H} = -t\sum_{\langle i,j\rangle}\left(\hat{b}_i^\dagger \hat{b}_j + \hat{b}_j^\dagger \hat{b}_i\right) + V\sum_{\langle i,j\rangle}\hat{n}_i\hat{n}_j, \qquad \hat{n}_i = \hat{b}_i^\dagger\hat{b}_i, \quad (1)$$

with the hard-core condition $\hat{n}_i \in \{0,1\}$, $\hat{b}_i^2 = (\hat{b}_i^\dagger)^2 = 0$. The sum is over nearest-neighbor bonds, counted once. We set $k_B = 1$, measure energy in units of t, and use periodic boundaries. Because $\hat{H}$ commutes with $\hat{N} = \sum_i \hat{n}_i$, it is block diagonal in particle number; each block is diagonalized on its own and the eigenvalues feed all three ensembles. In the occupation basis the hard-core operators carry no sign between sites, so a hop contributes exactly $-t$ and every block is a real symmetric matrix.

A basis state is a string of occupations stored as an integer, bit i holding n_i. The N-particle sector has dimension

$\binom{M}{N}$, and the sectors add up to 2^M. The canonical and grand-canonical sums run over the whole spectrum, not a few low-lying states, so we diagonalize each block densely rather than with a Lanczos method. The half-filled sector sets the cost: $\binom{16}{8} = 12\,870$ for $L = 4$, a few minutes on a workstation. We cache the spectra once.

The $L \times L$ tori are not the only periodic square lattices. Identifying sites that differ by integer combinations of two vectors $\mathbf{a}_1, \mathbf{a}_2$ gives a cluster of $M = |\det[\mathbf{a}_1\, \mathbf{a}_2]|$ sites; it is bipartite when both vectors have even $x + y$, and every site keeps four distinct neighbors when the cluster is not too small. We use $\mathbf{a}_1, \mathbf{a}_2 = (2,2), (2,-2)$; $(3,1), (1,-3)$; $(4,0), (0,4)$; $(3,3), (3,-3)$; $(4,2), (2,-4)$, for $M = 8, 10, 16, 18, 20$. All five are bipartite with coordination four, and all are even, so half filling is exact; $M = 16$ is the 4×4 torus again. The 2×2 torus fails the coordination test (it is a four-site ring) and 3×3 is not bipartite. To reach $M = 20$, whose half-filled sector has 184 756 states, we block-diagonalize each N-sector by momentum: projecting the sector Hamiltonian onto Bloch states $\sum_{\mathbf{r}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{T}_{\mathbf{r}} |a\rangle$ leaves M blocks of about $\binom{M}{N}/M$ states. Momenta related by a rotation or reflection that maps the cluster to itself share a spectrum, so one block per star suffices, and particle–hole symmetry, $E_{M-N} = E_N + V(2M - 4N)$, supplies the sectors with $N > M/2$. The largest dense block is then 9 278 states, and a full $M = 20$ spectrum takes about 15 minutes.

3 The three ensembles

Everything below starts from one object: the set of eigenvalues $E_{N,\alpha}$, where N labels the particle-number sector and α the level within it. The three ensembles are three ways of weighting that set.

3.1 Microcanonical

An isolated system with N particles and energy in a window $[E, E + \Delta E)$ visits each accessible level with equal probability. Counting them,

$$\Omega(E, N; \Delta E) = \#\{\alpha : E \le E_{N,\alpha} < E + \Delta E\}, \qquad S = \ln \Omega. \tag{2}$$

The first law without a volume term, $dE = T\, dS + \mu\, dN$, gives temperature and chemical potential as slopes of the entropy surface,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N, \qquad \mu = -T \left(\frac{\partial S}{\partial N}\right)_E. \tag{3}$$

Both slopes have to be taken as finite differences, since E is discrete and N is an integer. The window width ΔE is a modeling choice that does not go away: Ω is a histogram, and its bin size shows up in every quantity derived from it.

3.2 Canonical

Put the system in weak contact with a bath and hold the total energy fixed. The probability of the system sitting in level α is proportional to the number of bath states compatible with it, $\Omega_{\text{bath}}(E_{\text{tot}} - E_\alpha) = e^{S_{\text{bath}}(E_{\text{tot}} - E_\alpha)}$. The system energy is tiny against the bath, so expanding to first order,

$$S_{\text{bath}}(E_{\text{tot}} - E_\alpha) \simeq \text{const} - \beta E_\alpha, \qquad \beta = \frac{\partial S_{\text{bath}}}{\partial E} = \frac{1}{T}, \tag{4}$$

and the Boltzmann weight follows:

$$p_\alpha = \frac{e^{-\beta E_{N,\alpha}}}{Z_N}, \qquad Z_N(\beta) = \sum_\alpha e^{-\beta E_{N,\alpha}}. \tag{5}$$

Mean energy and its fluctuation are derivatives of $\ln Z_N$:

$$\langle E \rangle_N = -\frac{\partial \ln Z_N}{\partial \beta} = \sum_\alpha E_\alpha p_\alpha, \tag{6}$$

$$\mathrm{Var}(E) = \frac{\partial^2 \ln Z_N}{\partial \beta^2} = \langle E^2 \rangle - \langle E \rangle^2. \tag{7}$$

The heat capacity is fixed by the fluctuation, not by any separate measurement: differentiating $\langle E \rangle$ with respect to T and using $d\beta/dT = -\beta^2$,

$$C_V = \frac{\partial \langle E \rangle}{\partial T} = \beta^2\, \mathrm{Var}(E). \tag{8}$$

The Gibbs entropy $-\sum_\alpha p_\alpha \ln p_\alpha$ collapses to $S = \beta \langle E \rangle + \ln Z_N$, and $F_N = -T \ln Z_N = \langle E \rangle - TS$.

3.3 Grand canonical

Let the system trade particles with the reservoir as well. The same first-order expansion, now in both energy and number, with $\partial S_{\text{res}}/\partial N = -\beta\mu$, gives

$$p_{N,\alpha} = \frac{e^{-\beta(E_{N,\alpha} - \mu N)}}{\Xi}, \qquad \Xi(\beta, \mu) = \sum_N e^{\beta\mu N} Z_N(\beta). \tag{9}$$

The mean number and its fluctuation are μ-derivatives of $\ln \Xi$,

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu} = \sum_{N,\alpha} N\, p_{N,\alpha}, \qquad \frac{\partial \langle N \rangle}{\partial \mu} = \beta\, \mathrm{Var}(N), \tag{10}$$

the second being the compressibility sum rule, derived the same way as Eq. (8). Since $\mathrm{Var}(N) > 0$, $\langle N \rangle(\mu)$ increases monotonically, a fact we lean on when matching ensembles. The mean energy is $\langle E \rangle = -\partial_\beta \ln \Xi + \mu \langle N \rangle$.

3.4 When they agree, and by how much

The canonical partition function is a Laplace transform of the microcanonical count. Writing the level density as $\rho_N(E) = \sum_\alpha \delta(E - E_\alpha) \approx e^{S(E)}$,

$$Z_N(\beta) = \int dE\, \rho_N(E)\, e^{-\beta E} = \int dE\, e^{S(E) - \beta E}. \tag{11}$$

For large M the integral is dominated by the energy E^* where $S'(E^*) = \beta$—which is exactly the microcanonical relation Eq. (3) identifying the two temperatures. Expanding around the saddle, the canonical energy distribution is Gaussian with width $\sigma_E^2 = -1/S''(E^*)$. Entropy and energy are both extensive, so S' is intensive and $S'' \sim \mathcal{O}(1/M)$; hence $\sigma_E^2 \sim M$ and

$$\frac{\sigma_E}{|\langle E\rangle|} \sim \frac{\sqrt{M}}{M} = \frac{1}{\sqrt{M}}. \tag{12}$$

The same argument applied to $\Xi = \sum_N e^{\beta\mu N} Z_N$, now with a saddle in N, gives $\sigma_N/\langle N\rangle \sim 1/\sqrt{M}$. These two ratios are the small parameters of ensemble equivalence: when they are small the canonical energy and grand-canonical number distributions are sharp, and fixing T or μ is the same as fixing E or N. They are also pure spectral sums, with no differentiation, which is what makes them usable on a small discrete spectrum.

3.5 Why we do not read T off the entropy

Equation (3) is the honest definition of the microcanonical temperature, and on a small spectrum it is close to useless. $S(E) = \ln \Omega$ is a jagged histogram; its slope swings wildly, diverges wherever the matched energy lands in a gap, and changes sign past the middle of the band, where the density of states peaks and the lattice gas enters its negative-temperature regime. Figure 5 shows the damage. We tried surface, Gaussian-broadened, and volume (Gibbs) entropies; none rescues a single-energy thermometer at these sizes. So Eq. (3) stays in the paper as a diagnostic, with its ΔE dependence reported, and the quantitative claims rest on Eq. (12) and its number-channel twin.

To match the ensembles at a common operating point we fix N and β, take the canonical $\langle E\rangle_N(\beta)$, and solve $\langle N\rangle(\beta, \mu^*) = N$ for the chemical potential. Monotonicity of $\langle N\rangle(\mu)$ makes the root unique; Brent's method finds it to 10^{-10}. The energy per site compared across ensembles at that point, $|\langle E\rangle_{\text{grand}} - \langle E\rangle_{\text{can}}|/M$, is the third metric we track.

3.6 Four limits we can do by hand

The number channel has analytic limits that bracket what the numerics should do, and that let us separate finite-size effects from physics.

Infinite temperature. As $\beta \to 0$ at fixed filling $n = N/M$, every Fock state carries the same weight, so the grand-canonical N is binomial and

$$\frac{\sigma_N}{\langle N\rangle} \to \sqrt{\frac{1-n}{nM}}, \tag{13}$$

an exact $M^{-1/2}$ law for any t and V.

No hopping. Writing $\hat{n}_i = (1 + s_i)/2$ turns the $t = 0$ gas into an Ising model, $V\hat{n}_i\hat{n}_j \to \frac{V}{4} s_i s_j$ plus a field that vanishes at the particle–hole symmetric point $\mu = zV/2$ ($z = 4$), which is half filling. For $V < 0$ the coupling $J = |V|/4$ is ferromagnetic, and Onsager's solution of the square lattice [7] places the transition at

$$T_c = \frac{2J}{\ln(1+\sqrt{2})} \approx 0.567\,|V|. \tag{14}$$

Below T_c a lattice held at $\mu = zV/2$ is either mostly full or mostly empty: $N = M(1 \pm m_0)/2$, with Yang's spontaneous magnetization $m_0 = [1 - \sinh^{-4}(2J/T)]^{1/8}$ [8]. Then $\sigma_N/\langle N\rangle \to m_0$, a value that does not shrink with M at all. Phase separation shows up in this metric as an exponent near zero, not near $-1/2$.

Hopping: the XXZ map. With $\hat{b}_i^\dagger = \hat{S}_i^+$ and $\hat{n}_i = \hat{S}_i^z + \frac{1}{2}$, hard-core bosons are spin-$\frac{1}{2}$ operators [9] and

$$\hat{H} = \sum_{\langle i,j\rangle} \Big[-2t\big(\hat{S}_i^x \hat{S}_j^x + \hat{S}_i^y \hat{S}_j^y\big) + V \hat{S}_i^z \hat{S}_j^z \Big] + \text{field}. \tag{15}$$

For $V < 0$ this is a ferromagnet with $J_z = |V|$ and $J_{xy} = 2t$. It is easy-axis, and phase separates at finite temperature, only when $V < -2t$. For $-2t < V < 0$ it is easy-plane: a superfluid, with no phase separation. Hopping therefore pulls the true T_c below Eq. (14) and drives it to zero at $V = -2t$.

The Heisenberg point. At $V = -2t$ the model is exactly an SU(2) ferromagnet, so in 2D the Mermin–Wagner theorem rules out ordering at any $T > 0$ [10]. Its ground multiplet has total spin $M/2$ and one member in each N-sector. At $\mu = zV/2$ the $M+1$ members are degenerate, so as $T \to 0$ the grand ensemble is uniform on $N = 0, \ldots, M$ and, for even M,

$$\frac{\sigma_N}{\langle N\rangle} \to \sqrt{\frac{M+2}{3M}} \to \frac{1}{\sqrt{3}}. \tag{16}$$

This too is size independent, but it approaches its limit from above, so a fit on small lattices returns a small *negative* exponent.

4 Validation

Each check below is a regression test in the code. *(i)* Every sector block is symmetric to machine precision. *(ii)* Block dimensions are $\binom{M}{N}$ and sum to 2^M. *(iii)* A separately written full-space Hamiltonian is block diagonal in N, and the union of per-sector spectra equals its spectrum to 10^{-10}—particle number is conserved. *(iv)* For $V = 0$ on $L = 3, 4$ the one-particle sector reproduces the tight-binding band $-2t[\cos k_x + \cos k_y]$ to 10^{-15}; on the 2×2 plaquette it gives the four-site ring $\{-2t, 0, 0, 2t\}$. *(v)* $\operatorname{Tr} \hat{H}$ equals the summed interaction, since hopping is off diagonal. *(vi)* Ground-state energies from a sparse Lanczos solver agree with the dense spectra ($E_0/t = -10.347$ at $L = 3$, -18.000 at $L = 4$ for $V = 0$). *(vii)* $C_V = \beta^2 \mathrm{Var}(E)$ and $\partial_\mu \langle N\rangle = \beta \mathrm{Var}(N)$ match numerical derivatives of

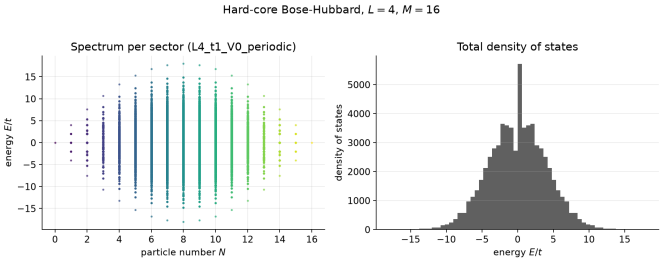


Figure 1: Eigenvalues per sector (left) and total density of states (right), $L = 4$, $V = 0$.

the means to 10^{-5}. *(viii)* The limits come out right: $S \to \ln(\text{ground degeneracy})$ and $C_V \to 0$ as $T \to 0$; $S \to \ln(\text{dim})$ and $\langle E \rangle \to$ the spectral mean as $T \to \infty$. *(ix)* The limits of Sec. 3.6 come out of the full spectra: Eq. (13) at $\beta t = 10^{-4}$ for repulsion, attraction, and the free gas, and Eq. (16) at $V = -2t$ to 10^{-9} (0.6124 at $M = 16$). *(x)* The symmetry-reduced spectra of the tilted clusters equal direct dense diagonalization to 10^{-13} in every sector for $M = 8, 10$, including the sectors obtained by particle–hole symmetry, and the $M = 16$ cluster reproduces the 4×4 torus. Bloch bases are orthonormal and their dimensions add up to $\binom{M}{N}$; the one-particle sector reproduces $-2t[\cos k_x + \cos k_y]$ on each cluster's allowed momenta.

5 Results

5.1 Spectrum, heat capacity, filling

The spectrum is symmetric about zero and peaked in the middle (Fig. 1)—a bounded band with a large zero-energy degeneracy, both generic to bipartite hopping. The heat capacity is a single broad bump that vanishes at both ends of the temperature range (Fig. 2); repulsion pushes the bump to higher T and attraction to lower, tracking the change in the characteristic energy scale. The grand-canonical filling passes through half-filling at $\mu = 0$ by particle–hole symmetry, and the $L = 3, 4$ curves have already nearly collapsed onto a common shape while $L = 2$ still stands apart (Fig. 3). Entropy per site climbs from zero at $T = 0$—a non-degenerate ground state—to the infinite-temperature ceiling $\ln(\text{dim})/M$, which itself creeps up toward $\ln 2$ as the lattice grows (Fig. 4).

5.2 The caloric curve, and the thermometry problem

Figure 5 is the clearest single picture of why the microcanonical temperature is not to be trusted here. The canonical $T(E)$, obtained by inverting the smooth $\langle E \rangle(T)$, rises cleanly through the lower half of the band. The finite-difference $1/(\partial S/\partial E)$ scatters around it below the band center and then goes negative above it. The negative values are not a bug—a bounded spectrum genuinely

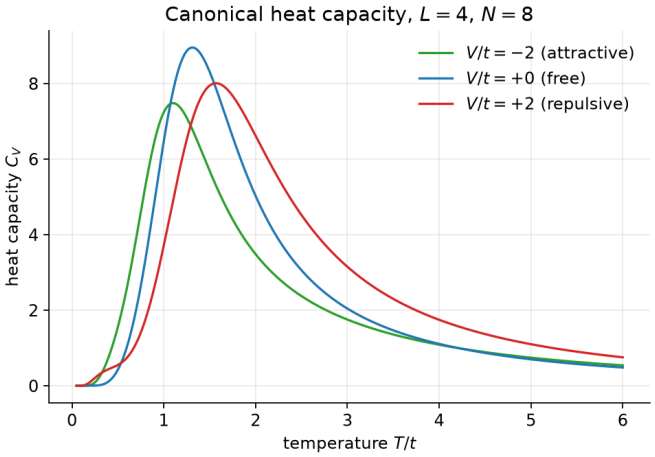


Figure 2: Canonical $C_V(T)$ at half filling ($L = 4$, $N = 8$).

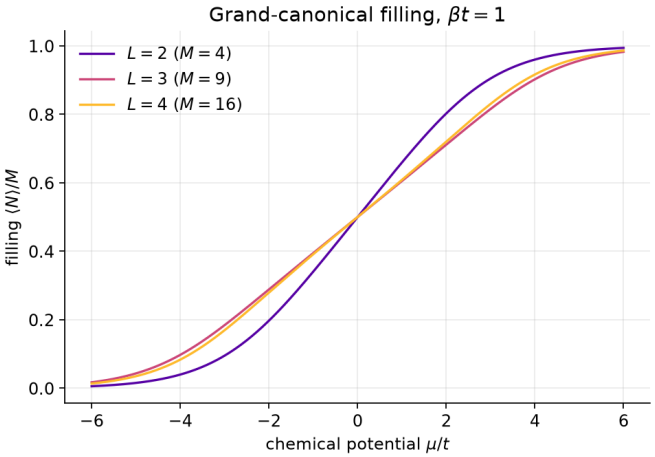


Figure 3: Grand-canonical filling versus μ/t at $\beta t = 1$.

has negative-temperature states—but they make the slope useless as a thermometer at a single energy.

5.3 Matching the ensembles, and the finite-size trend

Fix N and β, tune μ^* so $\langle N \rangle = N$, and the canonical and grand-canonical energies per site sit almost on top of each other, splitting by a small amount at intermediate temperature (Fig. 6). That gap, and the two fluctuation ratios, are what we follow against size.

Figure 7 and Table 1 collect the three metrics versus M for seven couplings. For the free gas and for repulsion all three fall with size; the free-gas $\sigma_N/\langle N \rangle$ and per-site gap follow clean power laws ($R^2 = 1.00$) with exponents between -0.6 and -0.9, straddling the $M^{-1/2}$ that Eq. (12) predicts. The energy ratio $\sigma_E/|\langle E \rangle|$ is the noisiest of the three, as expected—it feels the bounded band and the discreteness of the smallest lattice most strongly.

The attractive side behaves differently, and the difference is the main result. Reading down the $p(\sigma_N/\langle N \rangle)$ col-

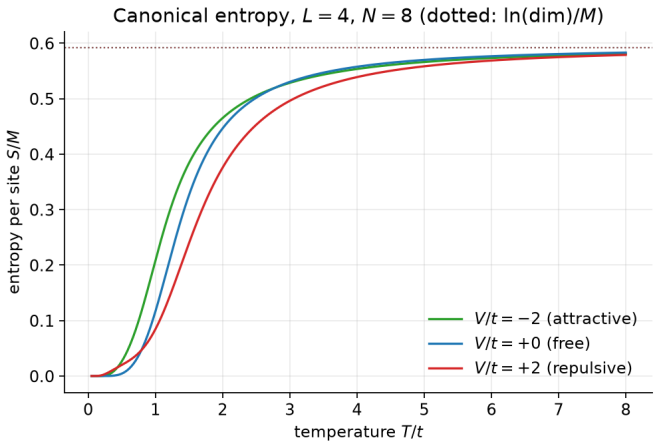


Figure 4: Entropy per site S/M; dotted lines mark $\ln(\text{dim})/M$.

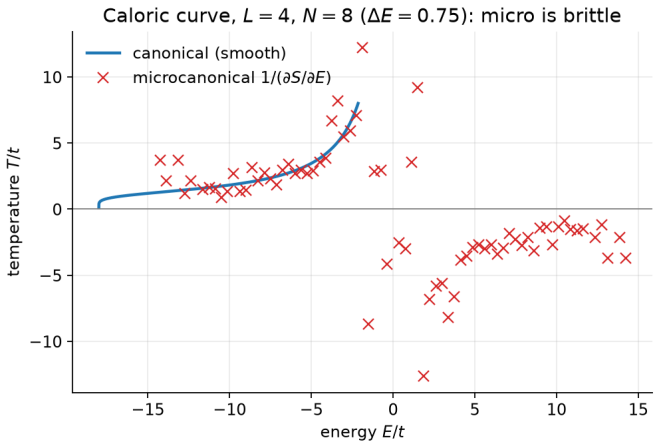


Figure 5: Caloric curve: smooth canonical $T(E)$ (line) against microcanonical finite differences (crosses), $L = 4$, $N = 8$, $\Delta E = 0.75$.

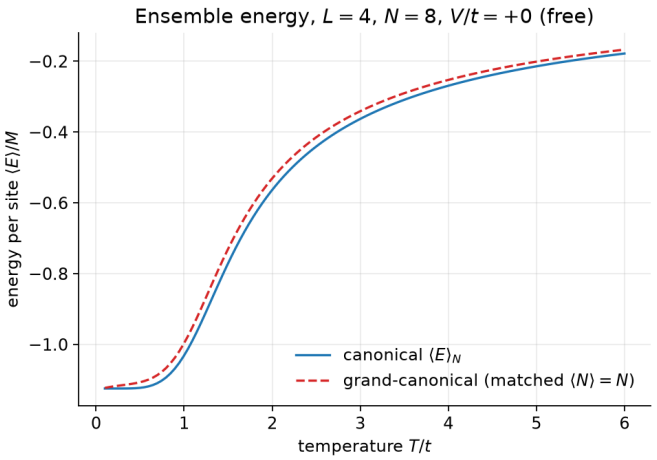


Figure 6: Canonical versus grand-canonical energy per site at matched N ($L = 4$, $N = 8$, $V = 0$).

Table 1: Fitted exponents p (metric $\propto M^p$, $M = 4, 9, 16$) and the number fluctuation on the largest lattice, $\beta t = 1$, half filling. The number-channel exponent rises steadily and turns positive at strong attraction.

V/t	$p(\sigma_E/\lvert\langle E\rangle\rvert)$	$p(\sigma_N/\langle N\rangle)$	$p(\Delta_E/M)$	$\sigma_N/\langle N\rangle$ (M=16)
-3	-0.92	$+0.13$	$+0.51$	0.962
-2	-0.88	-0.12	-0.59	0.553
-1	-0.91	-0.55	-0.37	0.240
0	-0.89	-0.64	-0.84	0.164
$+1$	-0.73	-0.65	-0.93	0.127
$+2$	-0.49	-0.69	-0.70	0.091
$+3$	-0.75	-1.10	-0.97	0.035

umn of Table 1, the decay exponent slides monotonically from -1.10 at $V/t = +3$, where repulsion restores equivalence fastest, through -0.64 for the free gas, up to $+0.13$ at $V/t = -3$. The sign flip is the point. At strong attraction the relative number fluctuation reaches 0.96 on the largest lattice and its exponent is positive, so the canonical-to-grand mismatch *grows* as the lattice grows. An attractive lattice gas wants to clump; the grand ensemble lets the particle number swing between the competing dilute and dense arrangements, and the fixed-N canonical ensemble cannot follow. The energy metrics keep shrinking for every V, so the failure lives specifically in the particle-number channel, which is where the instability lives too.

5.4 Interaction dependence

Plotting the metrics at $L = 4$ against V/t lays the trend bare (Fig. 8): $\sigma_N/\langle N\rangle$ falls monotonically from nearly 1 at $V/t = -3$ to a few percent at $V/t = +3$, crossing the energy ratio near $V = 0$. A finer sweep at fixed $L = 3$, where each coupling is almost free to compute, resolves the same curve over 33 points and shows it is smooth (Fig. 9).

5.5 The temperature axis

Everything above sits at $\beta t = 1$. Repeating the scaling fit at each of 40 temperatures from $T/t = 0.2$ to 10 gives the exponents as functions of T (Fig. 10); at $T = t$ they reproduce Table 1. At high temperature every coupling heads for Eq. (13): at $T = 10\,t$ the number exponent is between -0.43 and -0.54 for all seven couplings, against -0.49 for the binomial law fitted on the same three lattices.

The attractive anomaly is a low-temperature effect. At $V/t = -3$ the number exponent is positive only below $T \approx 1.9\,t$, peaking at $+0.14$ near $T = 1.15\,t$, and on 16 sites $\sigma_N/\langle N\rangle$ climbs to 1.00 as $T \to 0$ (Fig. 11). The temperature where the sign changes falls close to the classical $T_c = 1.70\,t$ of Eq. (14), and Yang's $m_0(T = t) = 0.994$ is close to the 0.962 we measure. Below the crossover the smaller lattices are rounded further below m_0 than the larger ones, so the ratio *rises* toward its size-independent limit and the fit returns a positive exponent. At the lowest temperatures all three lattices saturate and the exponent

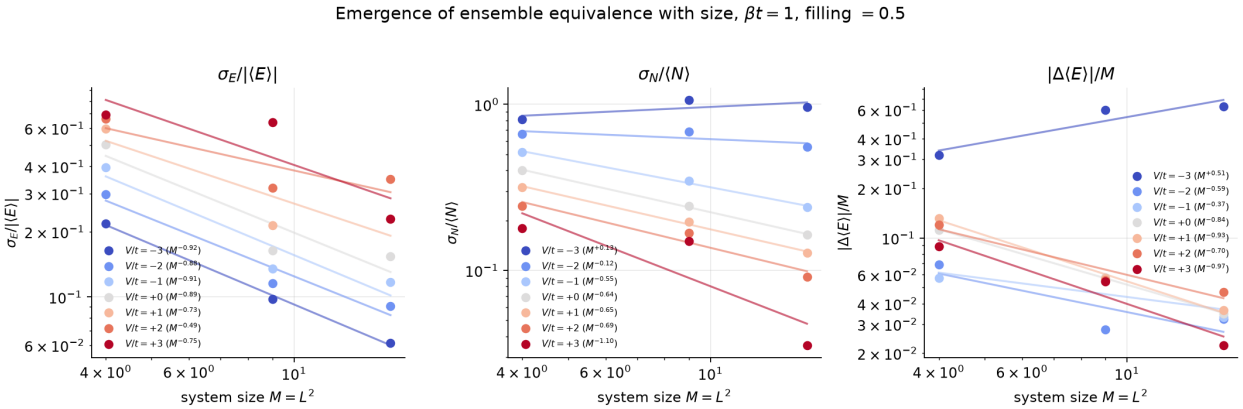


Figure 7: Finite-size scaling of $\sigma_E/|\langle E\rangle|$, $\sigma_N/\langle N\rangle$, and the per-site energy gap for seven couplings (attractive blue, repulsive red), $\beta t = 1$, half filling. Lines are power-law fits.

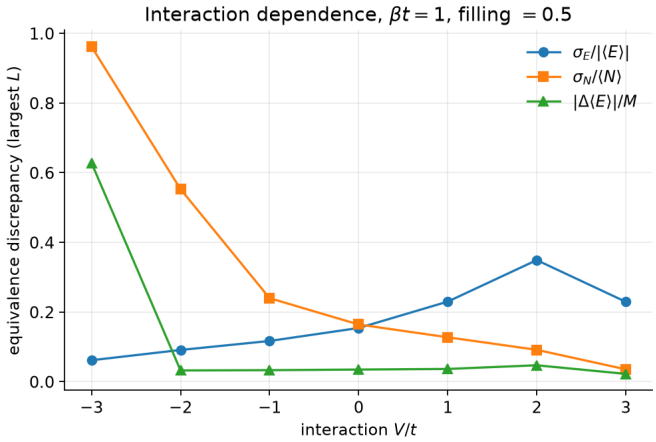


Figure 8: Equivalence metrics at $L = 4$ versus V/t, seven couplings.

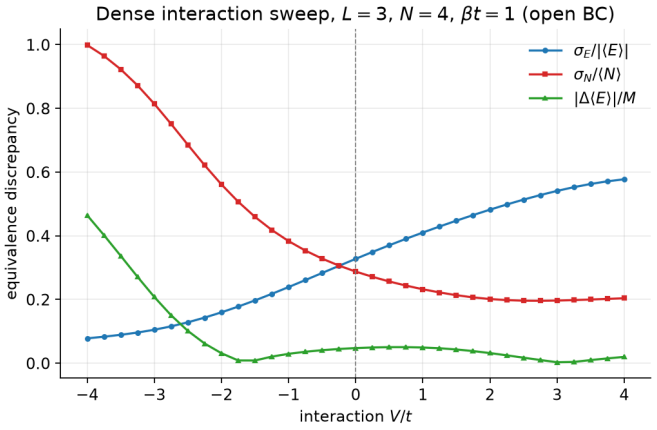


Figure 9: A denser sweep at fixed $L = 3$ (33 couplings, open BC, $\beta t = 1$): $\sigma_N/\langle N\rangle$ climbs smoothly toward unity with attraction.

drops back toward zero (+0.02 at $T = 0.2\,t$), as the Ising picture says it should.

$V/t = -2$ never turns positive. Its exponent flattens at about -0.1 below $T \approx t$, and its number fluctuation on 16 sites levels off at 0.612, which is Eq. (16) to nine digits. This is the Heisenberg point, the boundary of the anomaly. The classical Ising curves overstate the $V/t = -1$ and -2 fluctuations badly (Fig. 11), because at those couplings hopping is as strong as the attraction and the classical limit does not apply.

The free and repulsive gases converge at every temperature above $T \approx 0.5\,t$, with exponents between -0.43 and -0.70 for $V/t = -1, 0, +1, +2$, and steeper for $V/t = +3$, which approaches a gapped checkerboard solid at half filling. The map in Fig. 11 puts the zero contour on the attractive side next to the classical T_c line, and nowhere else above $T \approx 0.3\,t$.

5.6 One geometry: tilted clusters to 20 sites

Table 2 and Figs. 12–13 repeat the analysis on the five bipartite clusters. The number channel becomes clean. At high temperature every coupling gives -0.500, the binomial value, because every cluster sits exactly at half filling. At $T = t$ the free gas gives -0.47 and $V/t = \pm 1$ give -0.45, each with $R^2 = 1.000$; the free-gas value on the square tori, -0.64, was tilted by the four-site ring and the off-half-filled 3×3 lattice.

At $V/t = -3$ the exponent is +0.02 at $T = t$ and 0.00 below $T \approx 0.5\,t$, with $\sigma_N/\langle N\rangle = 0.965$ on 20 sites. It stays within 0.05 of zero up to $T \approx 1.5\,t$ and passes -0.25 near $T \approx 2.1\,t$, bracketing the classical $T_c = 1.70\,t$. This is the size-independent plateau that Sec. 3.6 predicts below T_c; the positive +0.13 of Table 1 was mostly a property of the three-geometry fit. $V/t = -2$ approaches the Heisenberg value from above, with exponent -0.07 at the lowest temperature and $\sigma_N/\langle N\rangle = 0.6055$ on 20 sites at $T = 0.2\,t$, equal to $\sqrt{22/60}$ from Eq. (16).

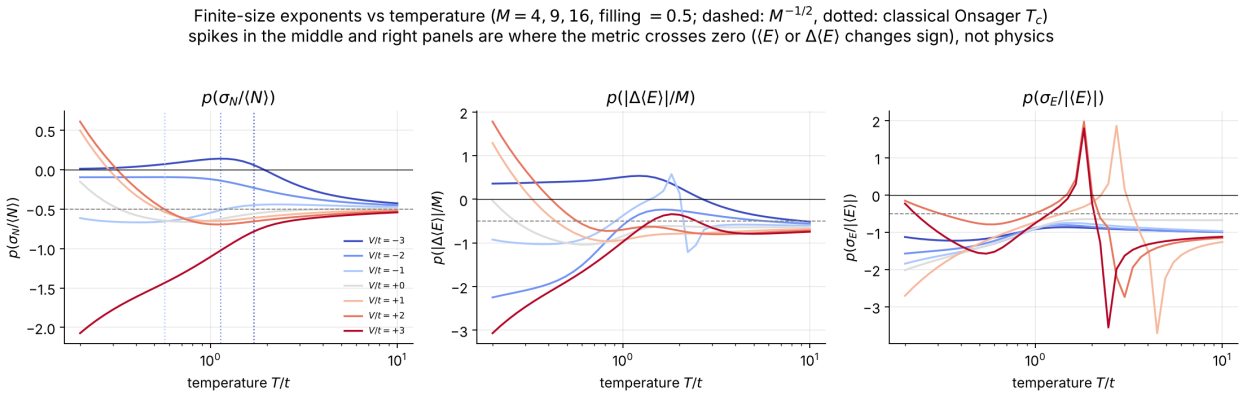


Figure 10: Fitted finite-size exponents ($M = 4, 9, 16$) of the three metrics versus temperature, seven couplings, half filling. Dashed: $M^{-1/2}$. Dotted: classical T_c of Eq. (14) for the attractive couplings. Spikes in the middle and right panels are zero crossings of $\Delta\langle E\rangle$ and $\langle E\rangle$, not physics.

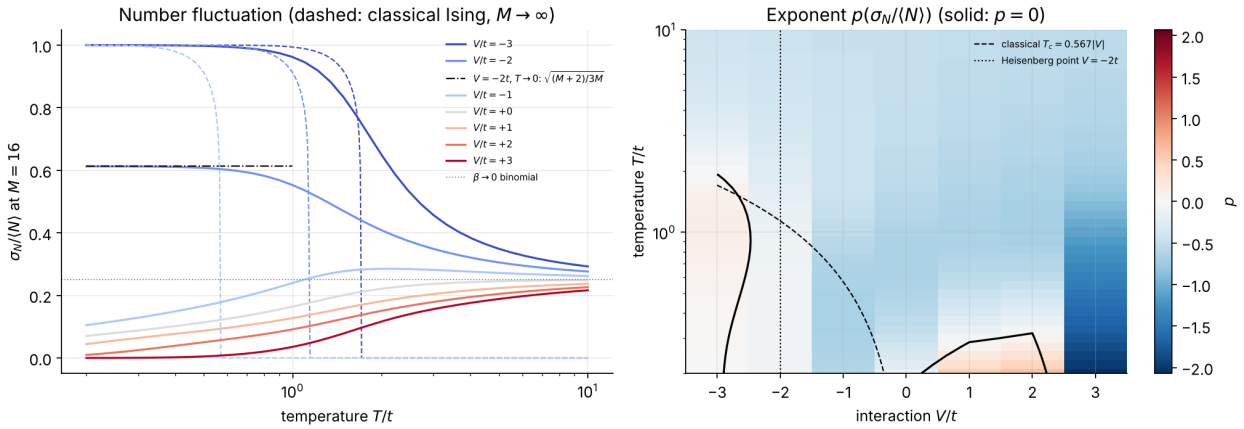


Figure 11: Left: $\sigma_N/\langle N\rangle$ on 16 sites versus T, against the $\beta \to 0$ binomial limit (dotted), the classical Ising $m_0(T)$ (dashed, $M \to \infty$), and the Heisenberg-point value Eq. (16) (dash-dot). Right: the number-channel exponent over the (V, T) plane, with its zero contour (solid), the classical T_c (dashed), and $V = -2t$ (dotted).

The energy channel changes more. The per-site gap falls as $M^{-0.9}$ to $M^{-1.0}$ for $|V| \le t$ at $T = t$ ($R^2 \ge 0.999$), and as $M^{-1.05}$ to $M^{-1.25}$ for every coupling at $T = 5\,t$. That is faster than $M^{-1/2}$, and it should be: canonical and grand-canonical means differ by an $\mathcal{O}(1)$ amount, so $\Delta\langle E\rangle/M \sim M^{-1}$. The relative energy fluctuation, by contrast, is nearly flat at $T = t$ (-0.06 for the free gas): the energy variance per site still grows from 8 to 20 sites, which cancels the $1/\sqrt{M}$. At $T = 10\,t$ its exponents lie between -0.46 and -0.72 for every coupling; the repulsive ones get there through zero-crossing spikes near $T \approx 2$–$3\,t$. The -0.89 of Table 1 came almost entirely from the four-site ring, where $\sigma_E/|\langle E\rangle| = 0.50$ against about 0.16 on every larger lattice.

Table 2: Fitted exponents at $\beta t = 1$, half filling, on the square tori ($M = 4, 9, 16$; Table 1) and on the bipartite tilted clusters ($M = 8, 10, 16, 18, 20$).

	$p(\sigma_N/\langle N\rangle)$			tilted
V/t	square	tilted	$p(\Delta_E/M)$	$\sigma_N/\langle N\rangle$ (M=20)
-3	$+0.13$	$+0.02$	$+0.10$	0.965
-2	-0.12	-0.19	$+0.50$	0.529
-1	-0.55	-0.45	-0.90	0.216
0	-0.64	-0.47	-0.94	0.148
$+1$	-0.65	-0.45	-0.98	0.115
$+2$	-0.69	-0.38	-0.37	0.084
$+3$	-1.10	-0.66	-0.11	0.030

5.7 What to distrust

Three sizes make for soft exponents, and the smallest point does the most to tilt a fit. It also happens to be the oddest lattice: the $L = 2$ torus de-duplicates to a single-bonded four-site ring instead of the true double-bonded two-torus, with a one-particle band $\{-2t, 0, 0, 2t\}$ rather than the $-4t$ bandwidth minimum. The $L = 3$ torus has its own quirk—an odd period makes it non-bipartite, so it loses the $E \leftrightarrow -E$ spectral symmetry that $L = 2$ and $L = 4$ keep, though the $N \leftrightarrow M - N$ symmetry that pins $\langle N\rangle = M/2$ at $\mu = 0$ survives at every size. The $L = 2, 3, 4$ sequence is therefore three somewhat different

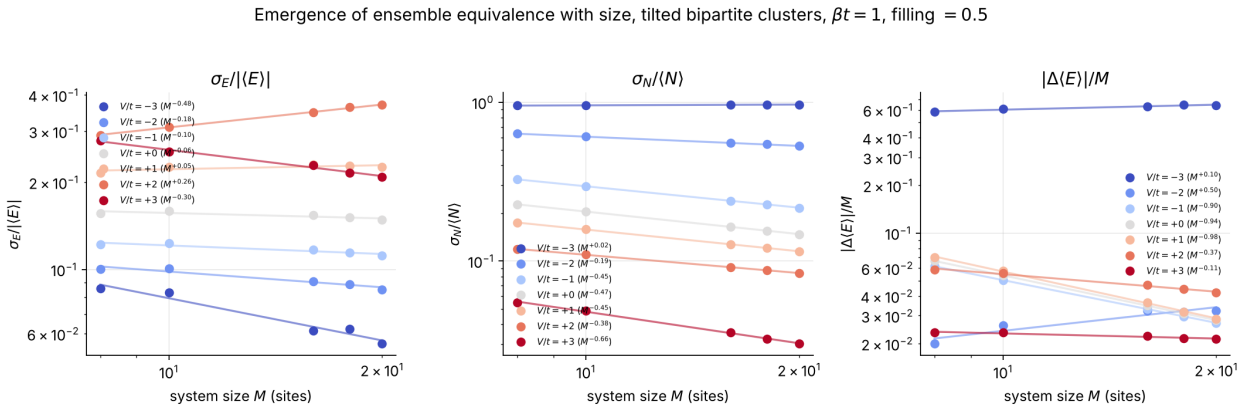


Figure 12: As Fig. 7, on the tilted bipartite clusters $M = 8, 10, 16, 18, 20$.

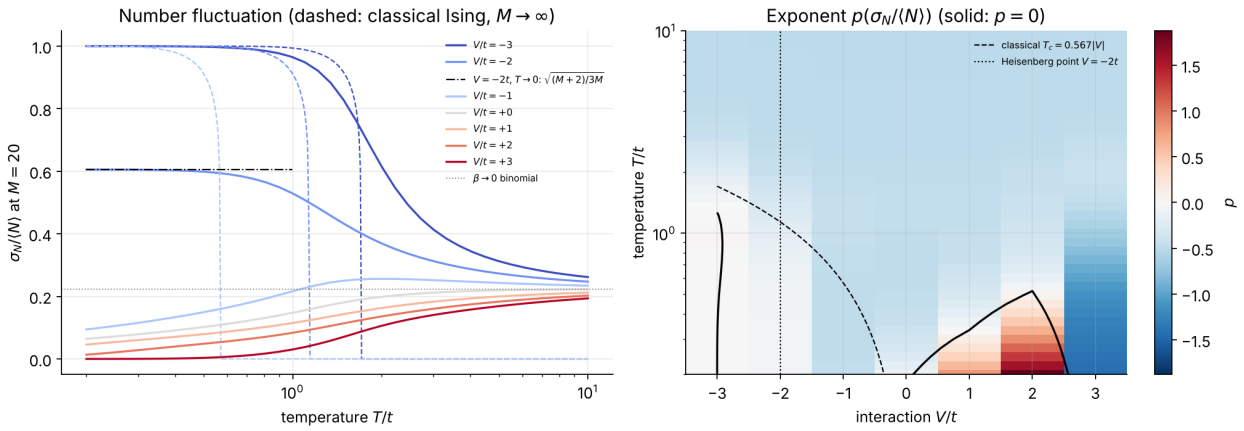


Figure 13: As Fig. 11, on the tilted clusters; left panel at $M = 20$. The attractive zero contour now hugs $V/t = -3$ below $T \approx 1.3\,t$: the exponent there is zero, not positive.

geometries stitched together, and some of the scatter in Table 1 is that stitching rather than physics. An open-boundary sequence, bipartite at every size, or an even-L series would be cleaner at the price of stronger edges. The stitching is visible even with no physics in it: the binomial law of Eq. (13), fitted on $M = 4$ and 9 alone (where $N = 4$, off half filling), gives an exponent of -0.36, not -0.5. The tilted clusters of Sec. 5.6 remove this problem, and where the two families disagree we trust the tilted one.

The temperature sweep adds two caveats. Below $T \approx 0.3\,t$ the $V/t = +1$ and $+2$ number exponents turn positive, to $+0.6$ on the square tori and $+1.9$ on the tilted clusters. The fluctuations there are a few percent at most and are set by each lattice's charge gap, which has not converged at these sizes; $V = +2t$ is also the onset of checkerboard order at half filling. That it survives on the bipartite clusters rules out frustration on 3×3 as the cause. It is a finite-size artifact, not an instability. Second, $\sigma_E/|\langle E\rangle|$ divides by an energy that passes through zero for the repulsive gas near $T \approx 2$–$3\,t$, which produces the spikes in Fig. 10; σ_E/M would avoid this. At low temperature the level spacing also suppresses the energy fluctuation more on smaller lattices, so the energy-ratio exponent is only meaningful above $T \approx 2\,t$ at these sizes.

6 Discussion

The choice to measure equivalence through fluctuations rather than through a matched temperature is not a workaround; it is the content of the saddle-point argument in Sec. III D. Equivalence *is* the statement that the Laplace transform relating the ensembles is dominated by a sharp saddle, and the width of that saddle is exactly $\sigma_E/|\langle E\rangle|$. The quantity we can compute cleanly is the quantity that was physically at issue all along.

The attractive anomaly reads naturally as the finite-size shadow of a first-order transition. Phase coexistence is the standard place where ensemble equivalence breaks even in the thermodynamic limit, because the microcanonical entropy loses concavity and the Maxwell construction separates the ensembles. We do not have a transition on 16 sites, but we have its precursor: a number distribution broad enough that fixing μ and fixing N stop being interchangeable, and broadening rather than sharpening as the lattice grows.

The temperature sweep and the limits of Sec. 3.6 make that reading concrete. The anomaly sits where the Ising map puts phase separation, below a temperature close to the classical T_c, and only on the easy-axis side of the XXZ map, $V < -2t$. It also clarifies what the signature is. Phase separation does not make the number fluctuation grow without bound; it makes it stop shrinking, an exponent near zero, and on small lattices the approach to that plateau can come from below or from above (a slightly negative exponent, $V/t = -2$). The tilted clusters confirm this: on one geometry the $V/t = -3$ exponent is zero to two decimals below $T \approx t$, and the positive value on the square tori was a fitting artifact. The sign alone is not the diagnostic. What separates the attractive side from the rest is that its exponent sits far from $-1/2$ at low temperature, where the free and repulsive exponents do not.

The next limit is size. A 26-site cluster has half-filled momentum blocks of about 400 000 states, out of reach for dense full-spectrum diagonalization; going further means giving up the full spectrum, for instance with a sign-free quantum Monte Carlo method for the canonical and grand ensembles, at the cost of the microcanonical one. At the sizes we have, the number channel is converged enough to be read quantitatively; the energy ratio is not, below $T \approx 2\,t$.

7 Conclusion

Built from one spectrum, the three ensembles of a small quantum lattice gas agree to a degree that improves with size for repulsive and non-interacting particles and fails to improve for strongly attractive ones, the failure confined to the particle-number channel where the gas would rather phase separate. Getting there required recognizing that the microcanonical thermometer breaks on small bounded spectra and measuring equivalence through its fluctuations instead, which is both what works numerically and what the saddle-point picture says to measure. A temperature sweep places the failure below the phase-separation temperature of the hopping-free gas and on the easy-axis side of the XXZ map, $V < -2t$, and the high-temperature and Heisenberg-point limits are reproduced exactly. On a single family of bipartite clusters up to 20 sites the free-gas number fluctuation falls as $M^{-0.47}$, and the strongly attractive one does not fall at all.

Author contributions. P. Das, A. Shakya, and D. Thakur conceived the study, developed the exact-diagonalization and ensemble framework, performed the calculations and analysis, and wrote the manuscript.

Reproducibility. Each cached spectrum and figure records the generating configuration (t, V, L, boundary condition, grids, ΔE). Spectra are built once and the analysis reads the cache; the validation checks of Sec. IV run as a test suite.

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